

Ernie Maldonado

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Summary

Data & ML Engineer with 10+ years building large-scale pipelines and production models for fraud, sanctions, and cybersecurity at Citigroup, Bank of America, and Certegy. Currently rebuilding the analytics layer – **Python** forecasting, metrics, and dashboards plus **Java/Spring Boot** services – for a ground-up rewrite consolidating 28 fraud applications into a single platform. Former part owner of Grupo BIT, a CPG analytics firm acquired by Teamcore (Accel-KKR) in 2024. Track record: 500K monthly sanctions alerts automated, check-fraud detection rates doubled, processing times cut 90%. Three graduate degrees; AWS Certified Cloud Practitioner, pursuing AWS Certified Data Engineer – Associate.

Core Skills

Python • SQL • PySpark • Hadoop/Hive • XGBoost • Scikit-Learn • AWS • Docker • FastAPI • Devin AI

Work Experience

Citigroup, Tampa FL **2018 – 2021, 2022 – 2023, 2024 – Present**
Internal Fraud Engineer *2024 – Present*

- Rebuild the analytics layer for a ground-up rewrite consolidating **28 applications** into a single platform: own the **Python** forecasting, metrics, and dashboard data end to end, and build backend services in **Java/Spring Boot**; use **Devin** (AI coding agent) to accelerate delivery.
- Mine Hadoop databases (transactions, user behavior, historical cases) to surface fraud patterns and translate findings into **PySpark/Hive** detection rules targeting anomalous velocity, high-risk access, and suspicious vendors.
- Collaborate with analysis teams to simulate, tune, and deploy alert logic; manage alert volume by focusing on recurring MOs that mirror confirmed past crimes.

Sanctions Machine Learning Engineer *2022 – 2023*

- Automated 500K monthly sanctions alerts by building end-to-end NLP pipelines (Random Forest, Gradient Boosting, XGBoost) with 50+ engineered features to match flagged customers against sanctioned-persons lists.
- Cut manual review volume by 50% by discovering Citi's internal transactions were being double-screened across two independent systems.
- Authored Model Risk Management documentation ensuring compliance with OCC 2011-12 and SR 11-7; produced architecture study showing a Hadoop migration would reduce processing time by 90%.
- Stack:** SQL, Pandas, Python, Scikit-Learn, PyTorch, XGBoost, FastAPI

Cybersecurity Big Data & ML Engineer *2018 – 2021*

- Built from scratch a production anomaly-detection system monitoring 300K employees across 5TB of weekly data using peer-normalized behavioral metrics and k-means clustering to surface insider threats for the Cyber Security Fusion Center.
- Engineered ingestion pipelines for 12+ data sources into a 270TB Hadoop cluster (Sqoop, Flume, Python parsers, Hive, Oozie) and built a monitoring system to ensure all feed flows remained current.
- Stack:** PySpark, PyTorch, Scikit-Learn, TensorFlow, Hive, Oozie, Hadoop, Python, SQL, Linux

ML Analytics Consultant & Partner **2023 – 2024**
Grupo BIT, Colombia *Contract*

- Delivered ML and analytics work for a CPG business-intelligence provider serving traditional retail channels (wholesalers, distributors, small retailers) across 18 countries.
- Supported the company through its sale process as part owner; acquired by **Teamcore** (Accel-KKR portfolio) in May 2024.

Consumer Protection Big Data & ML Consultant **2023**
Bank of America, Tampa FL *Contract*

- Audited the entire fraud **Hadoop** ecosystem and produced a data-driven benchmark report comparing current practices against industry standards across performance, reliability, and data quality.

Data Scientist – Fraud Detection ML **2021 – 2022**
Certegy Payments, Tampa FL *Hybrid*

- Doubled check-fraud detection rate at Walmart by developing Random Forest, Gradient Boosting, and Logistic Regression models; raised KS statistic from 0.05–0.3 to 0.6–0.8.
- Architected full ML pipeline: features built in Oracle SQL, models served via AWS API with real-time and batch scoring, routed to the correct model per transaction type.
- Stack:** SAS Viya, Oracle SQL, Oracle OML, Python, PyTorch, Scikit-Learn, TensorFlow, AWS

Data Scientist – Machine Learning **2016 – 2017**
Wellcare Health Plans (Centene), Tampa FL *On-Site*

- Reduced hospital readmissions by 60% using XGBoost and Scikit-Learn, directly limiting Medicare penalty exposure for the plan.
- Key contributor to a \$1M/year capital project coordinating Informatica, Hadoop, SAS, and Python workflows across external vendors and multiple business units.
- **Stack:** PySpark, Hive, Python, R, SQL, XGBoost, Scikit-Learn

Sr. Decision Support Analyst

2015 – 2016

Gateway HealthPlan (Highmark), Pittsburgh PA

On-Site

- Worked directly with the CFO to resolve reconciliation gaps between insurance claims and accounting systems via ETL improvements; built ARIMA cash-flow forecasting model in R; managed a team of three data engineers and supervised SAS platform administration.

Earlier Experience

2011 – 2014, 2021

Nielsen • UPMC Health Plan • University of Pittsburgh

- **Sr. Data Scientist**, Nielsen (contract, 2021) • **Database & ETL Analyst**, UPMC Health Plan (2012–2014) — Oracle SQL/SAS reporting layers; 2014 ACES Award, top 1% of employees • **Project Manager**, University of Pittsburgh (2011–2012) — two \$1.5M grant-funded research projects, 20+ staff.

Projects

NameGen: Generative AI Name Generator | Python, FastAPI, Docker

[Source Code](#)

- Generates novel names using directed-graph letter-transition probabilities sampled via multinomial distribution – the same statistical foundation underlying language model sampling.
- Deployed three ways: interactive CLI (with pickle-cached graph for speed), local FastAPI service, and Dockerized container, demonstrating a complete ML deployment lifecycle.

E-Commerce Fraud Detection | Python, XGBoost, Optuna, joblib

[Source Code](#)

- Three-model ensemble (Random Forest, Isolation Forest, XGBoost) applied to e-commerce fraud; XGBoost hyperparameters tuned with Optuna.
- Models serialized with joblib pipelines, ready for production deployment.

Education

Duquesne University, Pittsburgh PA USA

2008 - 2011

Masters in Computational Mathematics

3.9 GPA

KDI School, Seoul (Relocated to Sejong City), S. Korea

2004 - 2005

Masters in Public Policy

3.9 GPA

Universidad de los Andes, Bogota, Colombia

1996 - 2003

Masters in Economics (2003) & Bachelors in Economics (2001)

3.7 GPA

Certificates

AWS Certified Data Engineer – Associate (In Progress) • AWS Certified Cloud Practitioner (2022) • TensorFlow Specialization (2020) • Deep Learning Specialization (2018) • Machine Learning Specialization (2017) • Machine Learning Certificate (2015)

Technical Skills

Languages: Python, SQL, Java, R, SAS, Rust, Matlab

Big Data & Pipelines: Apache Hadoop, Hive, PySpark, Oozie, Impala, Sqoop, Flume

Cloud & Databases: AWS, Oracle DB, MS SQL Server, MySQL

Machine Learning: Scikit-Learn, PyTorch, TensorFlow, XGBoost, Optuna, SAS Viya, PySpark MLlib

GenAI & NLP: LLM APIs, AI coding agents (Devin), prompt engineering, NLP feature engineering, generative modeling

MLOps & Tools: FastAPI, Spring Boot, Docker, Flask, Git, GitHub, GitLab, BitBucket, Postman, Jira

Selected Publications

Participatory Assessment of the Health of Latino Immigrant Men in a Community – Journal of Immigrant and Minority Health (2013) • Bayesian Regression Inference Using a Normal Mixture Model – Masters Thesis, Duquesne University (2012) • FARC Terrorism in Colombia: A Clustering Analysis – CEDE Working Papers Series (2004)